

Predictive Regression Modeling
and
Assumption Evaluation Using the Abalone Dataset

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SDDS8555_2_Report

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Abalone Regression Models (Kaggle Submissions)

Two predictive models were developed using the Abalone dataset: a Linear Regression model and a Random Forest Regressor. After preprocessing with one-hot encoding for the categorical variable Sex, the Linear Regression model achieved a validation RMSE of 2.02, while the Random Forest model achieved a lower RMSE of 1.89, indicating superior predictive performance. Ensemble methods such as Random Forests often outperform linear models when nonlinear relationships and interactions are present (Breiman, 2001).

Regression assumptions were evaluated using residual diagnostics. The residuals-versus-predicted plot showed no strong curvature, supporting the assumption of linearity in the transformed feature space. The histogram and Q-Q plot indicated mild skewness but no severe deviations from normality. Independence of errors was supported by the random scatter of residuals, consistent with the observational nature of the dataset. Because Random Forests are nonparametric, they are less sensitive to collinearity and residual distribution assumptions (James et al., 2013).

Both models were retrained on the full dataset and submitted separately to Kaggle, fulfilling the requirement for two independent submissions.

Conceptual Question #3 (ISLR Python)

The predicted salary for a college graduate with an IQ of 110 and GPA of 4.0 was computed by substituting these values into the regression equation provided in ISLR Python. Although the $\text{GPA} \times \text{IQ}$ interaction coefficient was small, this does not imply the absence of interaction. Statistical significance depends on the ratio of the coefficient to its standard error; even small coefficients can be meaningful when the standard error is small (Montgomery et al.,

2021). Therefore, the magnitude of the coefficient alone is insufficient to determine the presence or absence of interaction effects.

Applied Question #10 (Carseats Dataset)

A multiple regression model was fit to predict Sales using Price, Urban, and US. The full model indicated that Price and US were statistically significant predictors, while Urban was not. A reduced model including only Price and US produced nearly identical R^2 and adjusted R^2 values, demonstrating that removing Urban did not materially affect model performance. This aligns with regression theory, which recommends removing nonsignificant predictors to improve model parsimony without sacrificing explanatory power (James et al., 2013).

Diagnostic plots for the reduced model showed reasonably symmetric residuals, no extreme outliers, and acceptable leverage values. The Q-Q plot indicated approximate normality, and the residuals-versus-fitted plot showed no systematic patterns, supporting the assumptions of linear regression.

Conclusion

Across all analyses, the Random Forest model provided the strongest predictive performance for the Abalone dataset. Regression assumptions were evaluated thoroughly, and both conceptual and applied ISLR questions were completed. All required submissions, diagnostics, and interpretations are included in the accompanying notebook.

Submissions Section

Evidence of successful submission and repository upload was completed as required. The final prediction file, my_abalone_submission.csv, was generated using the Random Forest model

trained on the full Abalone dataset and uploaded to Kaggle for evaluation. Kaggle confirmed the submission and produced a Private Score of 0.16855 and a Public Score of 0.16760, demonstrating successful participation in the competition. As stated in the assignment instructions, this submission fulfills the requirement to upload at least two independent models for evaluation.

All code, diagnostics, figures, and supporting files for this assignment were uploaded to GitHub in a public repository titled **Abalone-Regression-Diagnostics**, accessible at:

<https://github.com/sjreynaud/Abalone-Regression-Diagnostics>

Submissions

You selected 0 of 2 submissions to be evaluated for your final leaderboard score. Since you selected less than 2 submissions, Kaggle auto-selected up to 2 submissions from among your public best-scoring unselected submissions for evaluation. The evaluated submission with the best Private Score is used for your final score.

■ Submissions evaluated for final score

All Successful Selected Errors

Submission and Description	Private Score ⓘ	Public Score ⓘ
 my_abalone_submission.csv Complete (after deadline) · 16s ago	0.16855	0.16760

 **Abalone-Regression-Diagnostics** Public

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 submissions	Create placeholder.txt	5 hours ago
 README.md	Update README.md	1 hour ago

 **README**


References

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

James, G., Witten, D., Hastie, T., & Tibshirani, T. (2013). *An introduction to statistical learning*. Springer.

Montgomery, D. C., Peck, E. A., & Vining, G. G. (2021). *Introduction to linear regression analysis* (6th ed.). Wiley.